Exercise: Spatial pattern of trees

Task (1)

In this study involving ten plots with an area of 25 m² each, the focus is on analyzing tree spatial patterns and calculating stem density per hectare. The key aspects include determining the spatial pattern using Fisher's Index, a statistical method that assesses whether trees are randomly distributed, clumped, or uniformly spaced within the plots. Additionally, the study involves calculating stems per hectare by counting the number of trees in each plot and extrapolating to estimate the number of stems per hectare (10,000 m²). Furthermore, the spatial pattern is adjusted to a systematic distribution, aiming to achieve a more uniform and systematic arrangement of trees within each plot. This analysis is crucial for forest management and ecology, as it influences biodiversity, forest health, and resource management.

PLOT	N
1	12
2	5
3	2
4	2
5	12
6	17
7	3
8	1
9	12
10	3
Total number of stems	

In Step 1, the following calculations were carried out:

1. Mean number of stems per plot:
   Mean number of stems per plot=Total number of stemsNumber of plots=6.9Mean number of s tems per plot=Number of plotsTotal number of stems=6.9
2. Variance in the number of stems per plot: S n 2 =∑(X i −Mean) 2=29.69*Sn*2=∑(*Xi* −Mean)2=29.69

Here, the spatial pattern of trees based on Fisher's index was obtained, and the calculated Fisher's Index is approximately 1.604, which is greater than 1. Therefore, it indicates that the spatial pattern of trees implies a slightly uniform distribution.

In Step 2, stems per hectare were calculated:

Stems per hectare=Total number of stemsTotal area of sample plot×10000=2760 stemsStems per hect are=Total area of sample plotTotal number of stems×10000=2760 stems

As mentioned in Step 3, to modify the spatial pattern of trees to a systematic distribution, adjustments in the number of trees in each plot may be required. Systematic distribution implies a more uniform and regular spacing of trees, which can be achieved by either planting or removing trees to create a more consistent pattern across the plots.

plot	Original_N	Suggested_N	Mean_Stems_Per_Plot	Variance_Stems_Per_Plot
1	12	6.9	6.9	32.98888889
2	5	6.9	6.9	32.98888889
3	2	6.9	6.9	32.98888889
4	2	6.9	6.9	32.98888889
5	12	6.9	6.9	32.98888889
6	17	6.9	6.9	32.98888889
7	3	6.9	6.9	32.98888889
8	1	6.9	6.9	32.98888889
9	12	6.9	6.9	32.98888889
10	3	6.9	6.9	32.98888889
Fisher's Index	1.604	-	-	-
Stems per Hectare	2760	-	-	-

Task (2)

In this task, we aim to assess the spatial pattern of trees in a young stand to determine the necessity for thinning. The method employed involves measuring the distance between trees. This approach is crucial for understanding how trees are distributed across the area, whether they are too densely packed, or if there's enough space for healthy growth. In this task, there are two key aspects. The first is choosing the appropriate index. Therefore, Clark-Evans index was selected because it helps to analyze the spatial patterns of trees within a given area and determine whether the pattern of spacing among individuals is clustered, random, or regular spaced. The second part of the task is calculating the spatial pattern using the selected Index. This exercise is essential in forestry management, particularly in young stands where early interventions can significantly impact future health and productivity of the stand. By assessing the spatial pattern of trees, informed decisions about thinning practices to promote optimal growth conditions can be obtained.

point-tree	tree-tree
1.3	2.2
2.5	4.3
1.1	3.3
2.2	3.6
2.1	2.5
1.4	2.8
4.4	1.1
2.4	3.5
1.8	3.2
3	2.9
2.2	2.2
1.4	2.7
2.2	3.2
1.1	2.1
2.4	2.6
Average Distance	2.81
Density of trees	0.0015
Expetced Mean distance	12.91
Clark-Evans Index	0.218

Density of trees can be calculated as the number of trees divided by the area of the plot. If we consider a plot size of one hectare (10,000 m^2) and the number of trees equal to the number of rows in the data, the density is 0.0015 trees per square meter. The calculated R value, at 0.218 indicates a high level of clustering among the trees in young stand because it is significantly less than 1. This indicates that the trees are more densely packed than they would be in a random distribution. Due to the pronounced clustering suggested by this low R value, it might be necessary to consider thinning within the young stand to alleviate competition for resources and encourage better growth.

Task (3)

In this part, the objective is to determine whether the spatial distribution of infected trees in a forest is random using the chi-square test. The chi-square test is a statistical method employed to assess if there is a significant difference between expected and observed frequencies in one or more categories. This exercise holds significance in forest pathology and management as understanding the pattern of disease spread aids in formulating strategies for disease control and management.

2	4	6	3	2	2
2	4	0	0	4	6
2	4	0	2	2	3
2	1	4	0	0	0
6	6	3	4	3	0
6	6	2	2	2	6

Forest area

Number of plots	36
number of infected trees	101
Expected Count per Plot	2.81
Chi square statistic	51.91
critical value at 0.05 significant level	49.8

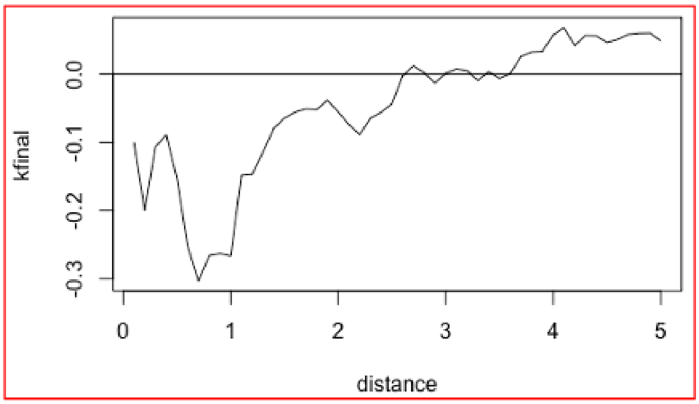
Chi square statistic $$\sum \frac{(O_i - E_i)^2}{E_i}$$

degree of freedom N - 1 = 35

In analyzing the spatial distribution of trees, we hypothesize that it is random (H0). Given the chi-square (χ^2) statistic of 51.91 surpasses the critical threshold of 49.80, we must reject the null hypothesis of randomness. Consequently, the spatial arrangement of infected trees in the forest appears to be non-random, possibly indicating a clustered distribution. This finding is critical for comprehending how infections spread among the trees and for guiding strategic forest management and intervention plans.

Task (4)

In this exercise, the objective is to analyze the spatial pattern of trees within a specified forest area covering *X* hectares. The analysis involves randomly placed plots, and the data utilized is derived from aerial images captured by drones employing an automatic tree detection algorithm. Ripley's L(t) function is implemented in R to calculate the spatial pattern of trees within the selected plots. Ripley's L(t) function is a spatial statistic used to assess patterns at various scales. This exercise is a component of spatial ecological analysis, providing valuable insights into the distribution of trees in a forest. Such analysis is crucial for understanding ecological dynamics, managing forest resources, and formulating effective conservation strategies.



The graph illustrates that at shorter distances, the tree spacing displays a higher level of regularity than what would be expected in a random distribution. This phenomenon may be attributed to factors such as competitive interactions, territorial behaviors, or other constraints. As the distance increases, this regularity diminishes, approaching a pattern consistent with random spatial distribution. Any deviation from this expected trend line implies non-randomness, suggesting either clustering or even spacing among the trees.

To extend these results to cover the entire forest area, comprehensive spatial data for all trees in the forest must be collected and prepared. This data should then be input into statistical software like R in the correct format for spatial analysis. Subsequently, point patterns and appropriate edge corrections should be analyzed using R software, employing packages like spatstat to ensure the analysis is not biased by boundary shapes.

The next steps involve computing and visualizing Ripley's K-function, determining it with an appropriate distance parameter. The resulting data is then visualized to assess the spatial distribution and compare it to theoretical models, helping classify it as random, clustered, or uniform. Statistical testing and interpretation are crucial components of the process. The observed K-function is statistically tested against a theoretical model to identify significant differences, and patterns are interpreted, taking into consideration ecological factors.

Task (5)

In this exercise, we are investigating the spatial autocorrelation of tree diameter within a forest using Moran's I statistic. Moran's I is a measure of spatial autocorrelation designed to assess whether similar values tend to occur near each other, or if there is no discernible pattern to the distribution of values in space. Additionally, we calculate Geary's C using the respective equation and draw interpretations based on the results.

This exercise is of paramount importance in the field of forest ecology and management. It contributes to a deeper understanding of the patterns of tree growth within a forest, providing insights that can be instrumental in making informed management decisions. These decisions may include actions such as thinning or implementing conservation efforts, optimizing the management strategies for the specific spatial patterns observed in tree diameters.

Moran's I Calculation:

$$\boxed{Moran\ I = \frac{n * \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(m_i - \bar{m}) * (m_j - \bar{m})}{2 * A \sum_{i=1}^{n} (m_i - \bar{m})^2}}$$

Consider two specific trees, Tree 1 and Tree 2, with the following information:

- Diameter of Tree 1 (mi) = 25
- Diameter of Tree 2 (mj) = 20
- Mean Diameter ($\bar{m}$) = 18.47
- Spatial Weight between Tree 1 and Tree 2 (Wij) = 1 if they are neighbors, 0 otherwise (from the adjacency matrix)

Substituting the values for our example: 1·(25−18.47)·(20−18.47)1·(25−18.47)·(20−18.47)

This evaluates to approximately 10.02.

Geary's C Calculation:

$$\boxed{Gearyn\ C = \frac{(n-1) * \sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(m_i - m_j)^2}{4 * A \sum_{i=1}^{n}(m_i - \bar{m})^2}}$$

Substituting the values for our example: 1·(25−20)21·(25−20)2

This evaluates to 25.

We have manually calculated Moran's I and Geary's C, and the obtained results are as follows:

1. Moran's I ~ 0.251
2. Geary's C ~ 0.599

Interpretation of Moran's I: The Moran's I statistic, ranging from -1 (indicating perfect dispersion) to +1 (indicating perfect correlation), with 0 denoting a random spatial arrangement, shows a moderate positive spatial autocorrelation in tree diameters with a value of approximately 0.251. This suggests that trees of similar diameters are more frequently found in proximity to one another than one would expect in a random spatial distribution.

Interpretation of Geary's C: Geary's C statistic, fluctuating between 0 (showing correlation) and 2 (showing dispersion), with 1 suggesting a random pattern, reveals a value of approximately 0.599. This indicates a moderate degree of spatial autocorrelation, aligning with the findings from Moran's I. It suggests that trees with similar diameters tend to be clustered together rather than randomly dispersed.